

《计算机网络》作业

第七次作业：（英文版教材第四章 18,20,22,24,39,41,42）

18.(4.3.5 Fast Ethernet) Ethernet frames must be at least 64 bytes long to ensure that the transmitter is still going in the event of a collision at the far end of the cable. Fast Ethernet has the same 64-byte minimum frame size but can get the bits out ten times faster. How is it possible to maintain the same minimum frame size?

20. (4.3.6 Gigabit Ethernet) How many frames per second can gigabit Ethernet handle? Think carefully and take into account all the relevant cases. Hint: the fact that it is gigabit Ethernet matters.

22. (4.4.3 The 802.11 MAC Sublayer Protocol) In Fig. 4-27, four stations, A, B, C, and D, are shown. Which of the last two stations do you think is closest to A and why?

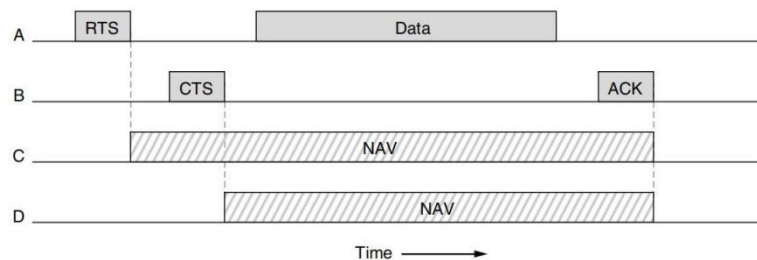
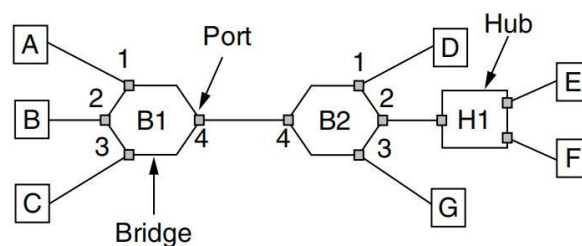


Figure 4-27. Virtual channel sensing using CSMA/CA.

24. (4.4.3 The 802.11 MAC Sublayer Protocol) A wireless LAN with one AP has 10 client stations. Four stations have data rates of 6 Mbps, four stations have data rates of 18 Mbps, and the last two stations have data rates of 54 Mbps. What is the data rate experienced by each station when all ten stations are sending data together, and

- (a) TXOP is not used?
- (b) TXOP is used?

39.(4.8.2 Learning Bridges) Consider the extended LAN connected using bridges B1 and B2 in Fig. 4-41(b). Suppose the hash tables in the two bridges are empty.



List all ports on which a packet will be forwarded for the following sequence of data transmissions and fill in the forwarding tables for bridge B1 and B2:

- (a) A sends a packet to C.
- (b) E sends a packet to F.
- (c) F sends a packet to E.
- (d) G sends a packet to E.
- (e) D sends a packet to A.
- (f) B sends a packet to F.

Forwarding Table of B1

Stations	Ports

Forwarding Table of B2

Stations	Ports

41. (4.8.5 Virtual LANs) To make VLANs work, configuration tables are needed in the bridges. What if the VLANs of Fig. 4-47 used hubs rather than bridges? Do the hubs need configuration tables, too? Why or why not?

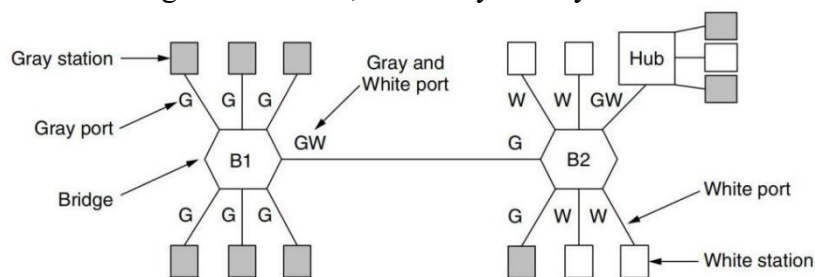


Figure 4-47. Two VLANs, gray and white, on a bridged LAN.

42. (4.8.5 Virtual LANs) In Fig. 4-48, the switch in the legacy end domain on the right is a VLAN-aware switch. Would it be possible to use a legacy switch there? If so, how would that work? If not, why not?

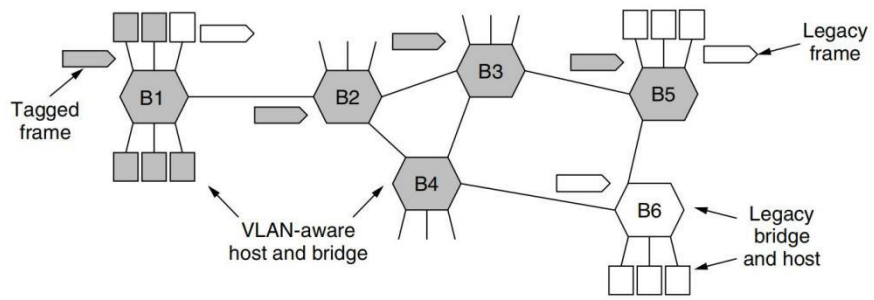


Figure 4-48. Bridged LAN that is only partly VLAN aware. The shaded symbols are VLAN aware. The empty ones are not.